

Ankit Tiwari

Data Science & ML Developer

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 Ayodhya, India

PROFILE

Data Science and Machine Learning developer with strong foundations in statistical learning, optimization, and neural networks. Builds ML systems and algorithms from first principles and develops applied AI solutions across multimodal learning and reinforcement learning.

PROJECTS

Customer Segmentation & Churn Prediction System

08/2025 – 09/2025

FastAPI, Scikit-learn, XGBoost, SHAP, MLflow

- Engineered an XGBoost-based customer churn prediction system achieving **ROC-AUC of 0.836** and **PR-AUC of 0.633**, with SHAP-driven insights for customer retention analysis.
- Deployed real-time churn scoring and customer segmentation APIs with **sub-100ms inference latency** for scalable prediction serving.
- Achieved **68.5% recall** in identifying churn customers using optimized threshold tuning and imbalance-aware training.
- Integrated SHAP explainability to identify key churn drivers such as tenure, contract type, and monthly charges, improving business interpretability.
- Developed data-driven retention targeting workflows by identifying high-risk customer segments for proactive churn mitigation strategies.

Healthcare RAG Intelligence System

10/2025 – 11/2025

Generative AI + LangChain + Gradio

- An AI system that helps hospitals instantly analyze patient reviews and answer operational queries using retrieved real-world feedback.
- Embeds hospital reviews, performs vector similarity search, and uses Gemini to generate context-aware answers with a Gradio chatbot interface.
- **Achieved 92%+ relevant response accuracy**, reduced review-analysis time by **85%**.

Real-Time Sign Language Translation System

03/2026 – 04/2026

Scikit-learn, FastAPI, Streamlit, Arduino (flex sensors)

- Achieved 25ms inference + 120ms E2E response on embedded edge hardware; built full Arduino→RandomForestClassifier→TTS pipeline with gesture-to-sentence mapping and confidence-filtered smoothing.
- Built a streaming pipeline (Arduino → sliding window → RandomForestClassifier → smoothing + confidence filtering → TTS + UI) with real-time API endpoints and gesture-to-sentence mapping
- Achieved **25ms** inference latency and **120ms** end-to-end response time with stable real-time predictions using smoothing and confidence thresholds.

Model-Based RL Agent with World Model Planning

08/2025 – Present

PyTorch, Gymnasium / DeepMind Control Suite, VAE, RSSM, LSTM/GRU, MCTS, CEM, TensorBoard

- Reduce sample inefficiency in reinforcement learning by enabling agents to learn environment dynamics and plan without excessive real-world interactions.
- Built an end-to-end MBRL pipeline combining VAE-based latent encoding, RSSM dynamics prediction, reward modeling, and planning (MCTS/CEM) to select optimal actions via imagined trajectories.
- Improved sample efficiency over model-free baselines (PPO/SAC) with reduced environment interactions and stable policy learning in continuous control tasks.

EDUCATION

Bachelor of Technology (Computer Engineering)

2022 – 2026

S.I.E.T Nilokheri, Haryana afw kurukshetra University, Kurukshetra 

CGPA: 7.4

SKILLS

Languages: Python, C, JavaScript, SQL

Libraries & Frameworks: Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn, Streamlit, TensorFlow, PyTorch, nltk, OpenCV, FastAPI, Next.js

Tools & Systems: Git, Docker, Linux, Jupyter Notebook

Technical Skills: LLMs/LLMs Fine Tuning, NLP, RAG pipelines, Vector Databases, Transformers(HuggingFace), CI/CD, REST APIs, model deployment

COURSES

Data Science Classroom

GeeksForGeeks

08/2025 – 11/2025 | Noida, India

CERTIFICATES

- Data Science & Machine Learning (GeeksForGeeks)
- Python Programming with DS&A(YBI Foundations) [↗](#)
- Next.js App Router Fundamentals (Vercel) [↗](#)
- React Foundations (Vercel) [↗](#)

INTERESTS

Quantitative Research, Systems Programming, Open source, Ethical Hacking (Nmap, BurpSuite)